

1] Explain the concept of JOIN in SQL. What is the difference between INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL OUTER JOIN?

Ans:

What is a JOIN

The JOIN clause in SQL is used to combine rows from two or more tables based on a related column between them. It allows querying data that spans across multiple tables by creating relationships between them.

Types of SQL JOINS

1. INNER JOIN:

Returns only the rows where there is a match in both tables.

Non-matching rows from either table are excluded.

Syntax:

SELECT columns

FROM table1

INNER JOIN table2

ON table1.common_column = table2.common_column;

2. LEFT JOIN (or LEFT OUTER JOIN:

Returns all rows from the left table and the matching rows from the right table.

If there is no match, NULL values are returned for columns from the right table.

Syntax:

SELECT columns

FROM table1

LEFT JOIN table2

ON table1.common_column = table2.common_column;

3. RIGHT JOIN (or RIGHT OUTER JOIN:

Returns all rows from the right table and the matching rows from the left table.

If there is no match, NULL values are returned for columns from the left table.

Syntax:

SELECT columns

FROM table1

RIGHT JOIN table2

ON table1.common_column = table2.common_column;

4. FULL OUTER JOIN:

Returns all rows when there is a match in either table.

Includes rows from both tables, with NULL values for non-matching rows in either table.

Syntax:

SELECT columns

FROM table1

FULL OUTER JOIN table2

ON table1.common_column = table2.common_column;